

IO183

Simulink Driver Block Manual

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1 Description

1.1 IO183

Introduction

The IO183 is a 16-bit analog module with either 8 single-ended or 4 differential analog inputs, 4 analog outputs and 16 digital TTL I/O lines configurable as inputs or outputs

- 8/4 analog input channels featuring:
 - Single-ended (8) or differential coupling (4)
 - Voltage range of ± 5 V, ± 10 V, 0–5 V and 0–10 V (applies to all inputs)
- 4 analog output channels featuring:
 - Voltage range of 0–2.5 V and 0–5 V (applies to all outputs)
- 16 digital I/O channels

Driver Blocks

- Setup
- Analog input
- Analog output
- Digital input
- Digital output

Pin Mapping

- Pin mapping for IO183

2 Pin Mappings

2.1 IO183 Pin Mapping

Connector A

Pin	Signal (Single-Ended Mode)	Signal (Differential Mode)
1	Analog Output 01	Analog Output 01
2	Analog Output 02	Analog Output 02
3	Analog Output 03	Analog Output 03
4	Analog Output 04	Analog Output 04
5	Analog Ground	Analog Ground
6	Analog Ground	Analog Ground
7	Analog Input 01	Analog Input 01 (-)
8	Analog Input 02	Analog Input 01 (+)
9	Analog Input 03	Analog Input 02 (-)
10	Analog Input 04	Analog Input 02 (+)
11	Analog Input 05	Analog Input 03 (-)
12	Analog Input 06	Analog Input 03 (+)
13	Analog Input 07	Analog Input 04 (-)
14	Analog Input 08	Analog Input 04 (+)
15	0 V	0 V
16	+5 V	+5 V
17	Analog Ground	Analog Ground

Note

0 V is internally connected to the ground of the target machine whereas analog ground is connected to the ground of the I/O module. 5 V are supplied by the real-time target machine, which is capable of providing 1 A continuous current. For more information, refer to the real-time target machine User Manual.

Connector B

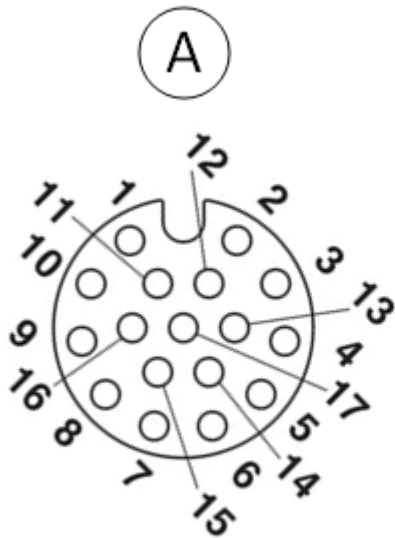
Pin	Signal
1	Digital I/O 01
2	Digital I/O 02
3	Digital I/O 03
4	Digital I/O 04
5	Digital I/O 05
6	Digital I/O 06
7	Digital I/O 07
8	Digital I/O 08
9	Digital I/O 09

Pin	Signal
10	Digital I/O 10
11	Digital I/O 11
12	Digital I/O 12
13	Digital I/O 13
14	Digital I/O 14
15	Digital I/O 15
16	Digital I/O 16 (+3.3 V optionally configurable by Speedgoat, see Note)
17	Digital Ground

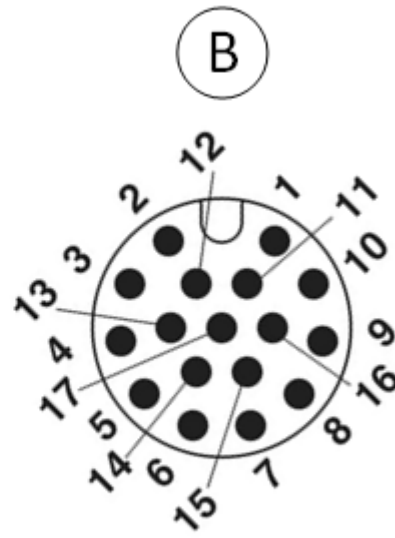
Note

Per default Digital I/O 16 is set a standard digital line. Optionally Speedgoat can set this line as a 3.3 V source during assembly (to be defined prior to order).

Pinout M12 Connector



17-pin M12 connector female



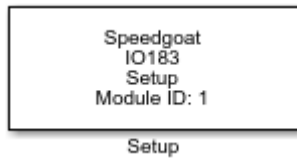
17-pin M12 connector male

3 Driver Blocks

3.1 IO183 - Setup

Library

Simulink Real-Time - Speedgoat



Description

Your model can contain only one Setup block for each I/O module in your target machine.

All the supported I/O types for this module can be configured in the Setup block dialog box. This block will therefore have an impact on all the other driver blocks (Analog input, Analog output, Digital input and Digital output).

Ports

This driver block has no input or output ports.

Parameters

The four tabs of the setup parameter dialog box contain the IO183 I/O module configuration settings.

Main Tab

Module ID

A unique module ID must be used for each type of I/O module Setup block in your model.

The module ID has two functions:

- It defines the logical connection to link the I/O module driver blocks with each other
- It also has an impact on the PCI slot auto-search feature: if only one I/O module is installed, the module ID must be set to 1. If multiple modules are installed, it must be in the range 1:n. Not all the I/O modules installed in the target machine need be used

PCI Slot (-1: autosearch)

There are two approaches for mapping the blocks to a specific I/O module installed in the target machine. All modules of the same kind must be configured using the same method.

- Auto-Search: the default value "-1" allows the real-time target machine to auto-search for the I/O module. The priority is defined by the module ID (first vector element for a multinode). The module with the lowest PCI bus/slot number is assigned to module ID 1 (or [1 x]) and the next module found is assigned to module ID 2 (or [2 x]), and so on
- Explicit Addressing: the location of the module inside the target machine can be explicitly defined by using the "[BusNumber, SlotNumber]" format. To determine the bus number and the PCI slot number, type "speedgoat_io" in the MATLAB command window

Simultaneous Analog Outputs

Enable or disable the simultaneous channel update. When enabled, all the analog outputs are updated simultaneously. This increases the task execution time, as the module waits for all the data to be transferred from every active channel before initiating the analog output update. When disabled, each channel is updated immediately after the data transfer.

Analog Input Tab**Coupling**

Select either Single-Ended or Differential.

Input Range

Select the voltage range for the input channels: ± 5 V, ± 10 V, 0–5 V, 0–10 V.

Scan Start Channel

Select the first active channel. The module reads every channel from Scan Start Channel to Scan Stop Channel. In differential mode the highest channel count is 4.

Scan Stop Channel

Select the last active channel.

Analog Output Tab**Active Channels**

Select the active input/output channels in a vector. The order inside the vector defines the input port order of the input/output block; for example, to only use a certain number of channels in a specific order, type "[2 1 4 3]", and to use all the channels in a straight order type "1:8".

Output Range

Select the voltage range for all output channels: 0–2.5 V or 0–5 V

Initial Values

Define the initial signal level present on the outputs after the model has downloaded. The values can be set individually by entering a vector: the value at a certain position in the Initial Values vector is applied to the channel as defined in the Active Channels vector. A scalar value applies to all the channels; for example, for individual values, type "[1 1.5 0 2.5]", and to set all channels to zero, type "0".

Reset to Initial Values

Define whether the initial values are also applied once the model has stopped. The behavior can be set individually for each channel using a vector. A scalar value applies to all the channels; for example, for individual values, type "[1 0 0 1]", and to set all channels to zero, type "0". "1" means use the Initial Value and "0" means keep the latest value.

Digital I/O Tab**Active Output Channels**

Select the active output channels in a vector. The order inside the vector defines the input port order of the output block. A defined number of channels can be selected using square brackets, for example, [1 4 6]. A sequence of channels can be selected using a colon, for example, 1:7.

Initial Values

Define the initial signal level present on the outputs after the model has downloaded. The values can be set individually by entering a vector: the value at a certain position in the Initial Values vector is applied to the channel as defined in the enabled channels. A scalar value applies to all the channels; for example, for individual values, type "[1 0 0 1]", and to set all channels to zero, type "0".

Reset to Initial Values

Define whether the initial values are also applied once the model has stopped. The behavior can be set individually for each channel using a vector. A scalar value applies to all the channels; for example, for individual values, type "[1 0 0 1]", and to set all channels to zero, type "0". "1" means use the Initial Value and "0" means keep the latest value.

Active Input Channels

Select the active input channels in a vector. Check the active output channels to avoid conflicts.

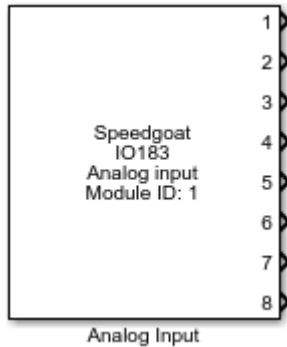
Pull Resistors - Front Connector

Select the reference voltage for the front I/O pull resistor: "Pull-up 3.3 V" or "Pull-down". The resistance value is 22 kΩ.

3.2 IO183 - Analog input

Library

Simulink Real-Time - Speedgoat



Description

Your model can only contain one of these blocks for each I/O module in your target machine.

Ports

This driver block has n double-type scalar output ports, where n is the number of selected analog input channels. Each port (signal) returns the analog value in volts.

Outputs

X (X is the analog channel number)

The value from analog channel X in volts.

Data Type: `double`

Parameters

The configuration details are all defined in the setup block. They are grayed-out because they are shown here just for reference.

Module ID

This ID defines the link to the corresponding Setup block.

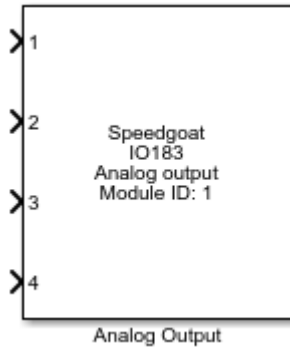
Sample Time

Defines the base sample time at which this driver block gets its sample hit. This parameter can also be set to -1 for inherited sample time. The units are in seconds.

3.3 IO183 - Analog output

Library

Simulink Real-Time - Speedgoat



Description

Your model can only contain one of these blocks for each I/O module in your target machine.

Ports

This driver block has n double-type scalar input ports, where n is the number of selected analog output channels. Each port (signal) accepts the analog value in volts to be output.

Inputs

X (X is the analog channel number)

The value to be output from analog channel X in volts.

Data Type: `double`

Parameters

The configuration details are all defined in the setup block. They are grayed-out because they are shown here just for reference.

Module ID

This ID defines the link to the corresponding Setup block.

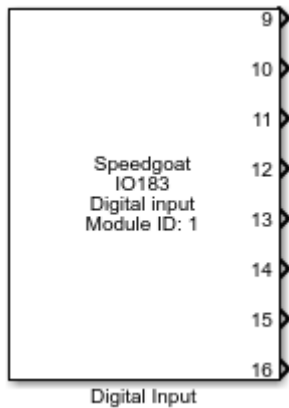
Sample Time

Defines the base sample time at which this driver block gets its sample hit. This parameter can also be set to -1 for inherited sample time. The units are in seconds.

3.4 IO183 - Digital input

Library

Simulink Real-Time - Speedgoat



Description

Your model can only contain one of these blocks for each I/O module in your target machine.

Ports

This driver block has n double-type scalar output ports, where n is the number of selected digital input channels. Each port (signal) returns the read digital value.

Outputs

X (X is the digital input channel number)

The following values are obtained from digital input channel X:

- 0 - TTL low
- 1 - TTL high

Data Type: double

Parameters

The configuration details are all defined in the setup block. They are grayed-out because they are shown here just for reference.

Module ID

This ID defines the link to the corresponding Setup block.

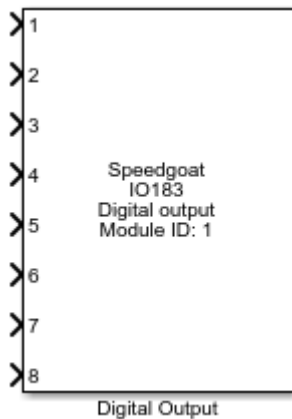
Sample Time

Defines the base sample time at which this driver block gets its sample hit. This parameter can also be set to -1 for inherited sample time. The units are in seconds.

3.5 IO183 - Digital output

Library

Simulink Real-Time - Speedgoat



Description

Your model can only contain one of these blocks for each I/O module in your target machine.

Ports

This driver block has n double-type scalar input ports, where n is the number of selected digital output channels.

Inputs

X (X is the digital output channel number)

Use the following values to set digital output channel X:

- 0 - TTL low
- 1 - TTL high

Data Type: double

Parameters

The configuration details are all defined in the setup block. They are grayed-out because they are shown here just for reference.

Module ID

This ID defines the link to the corresponding Setup block.

Sample Time

Defines the base sample time at which this driver block gets its sample hit. This parameter can also be set to -1 for inherited sample time. The units are in seconds.